Unit - I Operating System services and components

Q. Define real time operating system. List its any four applications of it. [2 m]

Ans.

A Real Time Operating System (RTOS) is an operating system that processes data and gives response within a fixed and small quantum of time.

Types of real time operating system

1. Hard real-time

2. Soft real-time

Applications:

1. Flight Control System

2. Simulations

3. Industrial control

4. Military applications

Q. Explain any 4 services provided by OS. [4 m]

Ans.

1. User Interface (UI)

Operating system provides an interface through which users can communicate with the computer system.

Types of user interface:

1. **Command Line Interface (CLI)**
2. **Batch Interface**
3. **Graphical User Interface (GUI)**

Example: Windows uses GUI while DOS uses CLI.

2. Program Execution

Operating system provides an environment to execute programs conveniently.

It performs tasks such as:

- Loading program into memory
- CPU scheduling
- Memory allocation and deallocation
- Terminating program after execution

It also handles abnormal termination by displaying error messages.

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3. I/O Operations

Operating system manages input and output operations of devices.

It provides services for devices such as:

- Keyboard
- Mouse
- Printer
- Scanner
- Hard disk

This makes communication between hardware devices and programs easier.

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4. File System Manipulation

Operating system manages files and directories stored in secondary storage.

It provides operations such as:

- Creating files
- Opening files
- Reading/Writing files

- Saving files
- Deleting files

Thus, OS makes file handling simple and efficient.

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5. Communication

Operating system allows communication between processes.

Communication can be done using:

- Shared memory
- Message passing

This helps processes exchange information efficiently.

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6. Error Detection

Operating system continuously checks the system for errors such as:

- Memory error
- Hardware failure
- Network failure
- Printer error

It takes suitable action to maintain proper system operation.

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7. Resource Allocation

Operating system allocates resources among different processes.

Resources include:

- CPU
- Main memory
- File storage
- I/O devices

It ensures efficient utilization of system resources.

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8. Accounting

Operating system keeps track of resource usage by users and processes.

It records:

- CPU usage
- Memory usage
- Disk usage

This information is useful for system management and billing.

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9. Protection and Security

Operating system protects data and resources from unauthorized access.

Security is provided using:

- User authentication
- Passwords
- Access control

Protection prevents one process from interfering with another process.

Q. Enlist types of operating system. Explain multiprogramming OS in detail. [4 m]

Ans.

Types of operating system

1.Batch Systems

2.Multiprogramming

3.Multitasking

4.Time-Sharing Systems

5.Desktop Systems

6.Distributed system

7.Clustered system

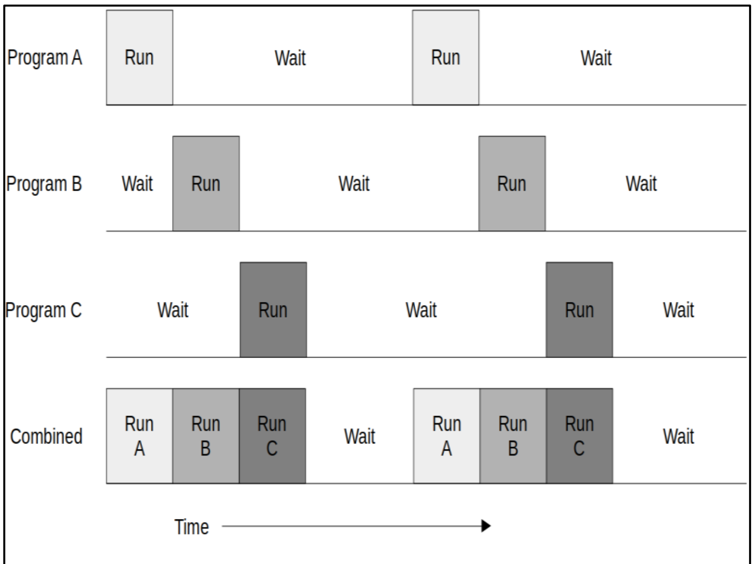
8. Real Time system

Multiprogramming:

- In multiprogramming, more than one program lies in the memory.
- The scheduler selects the jobs to be placed in ready queue from a number of programs.
- The ready queue is placed in memory and the existence of more than one program in main memory is known as multiprogramming.
- Since there is only one processor, there multiple programs cannot be executed at a time.
- Instead the operating system executes part of one program, then the part of another and so on.

Example:

user can open word, excel, access and other applications in a system.



Multiprogramming with Three Programs

Operating System
Program 1
Program 2
Program 3

Q. List components of OS. Explain process management in detail. [4 m]

Ans.

List of System Components:

1. Process management

2. Main memory management

3. File management

4. I/O system management

5. Secondary storage management

Process Management

The operating system manages many kinds of activities ranging from user programs to system programs such as printer spooler, name server and file server.

Each of these activities is encapsulated in a **process**.

- A process includes the complete execution context such as:
  - program code
  - data
  - program counter (PC)
  - CPU registers
  - operating system resources used during execution.
- The basic unit of software that the operating system schedules for CPU execution is a **process** or **thread**, depending on the operating system.
- A process is not just an application. An application may create multiple processes for different tasks.
- For example, an application like a word processor or browser may start additional processes for:

  - communication
  - printing
  - file handling
  - networking
- Some processes run in the background without direct interaction with the user.
- A process is software that performs a specific task and can be controlled by:
  - user
  - other applications
  - operating system
- The operating system controls and schedules processes for execution by the CPU.
- In a multitasking system, the operating system switches the CPU rapidly between processes for smooth execution.

Major Activities of Process Management

The five major activities of an operating system related to process management are:

1. Creation and deletion of user and system processes
2. Suspension and resumption of processes
3. Providing mechanism for process synchronization
4. Providing mechanism for process communication
5. Providing mechanism for deadlock handling

Q. What is purpose of system call? State two system calls with their functions. [4 m]

Ans.

Purpose of System Call

A system call provides an interface between a running user program and the operating system.

It allows user programs to access services provided by the operating system such as:

- process management
- file handling
- device management
- communication

System calls are procedures generally written using:

- C
- C++
- Assembly language instructions

Each operating system has its own set of system calls, and every system call is associated with a unique number that identifies it.

Types of System Calls

1. Process Control System Calls

These system calls are related to process execution and management.

A process must be loaded into memory before execution. During execution, the operating system may create, terminate, suspend or resume processes.

Functions

- create and terminate process

- load and execute process
- wait for event or time
- allocate and free memory
- get and set process attributes

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2. File Management System Calls

These system calls are used for performing operations on files and directories.

The operating system allows users to create, delete, open, close, read and write files.

File attributes include:

- file name
- file type
- file size
- protection information

System calls help in accessing and modifying these file attributes.

Functions

- create and delete files
- open and close files
- read, write and reposition files
- get and set file attributes

Examples

- open()
- read()
- write()
- close()

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3. Device Management System Calls

While executing, a process may require devices such as:

- printer
- disk drive
- keyboard

These system calls help in managing hardware devices.

Functions

- request and release device
- read and write device
- get and set device attributes
- attach or detach devices

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4. Information Maintenance System Calls

These system calls are used to exchange information between user programs and the operating system.

The operating system maintains information such as:

- current date and time
- memory status
- process information

Functions

- get/set time and date
- get/set system data
- get/set process attributes
- get/set device attributes

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5. Communication System Calls

Processes in the system communicate with each other using:

- message passing
- shared memory

Communication system calls establish connection between processes for exchanging information.

Functions

- create/delete communication connection
- send and receive messages
- transfer status information
- attach/detach remote devices

Q. Enlist the operating system tools. Explain any two in detail. [4 m]

Ans.

Operating System Tools

Following are the operating system tools:

1. User Management
2. Security Policy
3. Device Management
4. Performance Monitor
5. Task Scheduler

A) User Management

User management includes creating, modifying and deleting users in the system.

In Linux, user management can be performed using command line tools such as:

- useradd
- userdel
- usermod
- passwd

These commands are mainly used by system administrators.

1. useradd Command

This command is used to create a new user account.

Syntax

useradd -m -d /home/<userName> -c "<userName>" <userName>

Example

useradd -m -d /home/xyz -c "xyz" xyz

The file /etc/default/useradd contains default options for user creation.

Display Default Settings

useradd -D

2. userdel Command

This command is used to delete a user account.

Syntax

userdel -r <userName>

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3. usermod Command

This command is used to modify properties of an existing user.

Syntax

usermod -c '<newName>' <oldName>

Example

usermod -c 'vppoly' john

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4. passwd Command

This command is used to set or change the password of a user.

Syntax

passwd <userName>

Example

passwd vppoly

B) Device Management

- Device management is the process of managing physical and virtual devices connected to the computer system.
- In Linux, all device files are stored in the /dev directory, which is an important part of the root filesystem.
- These device files are required during the booting process.

Example

ls -l /dev

This command displays the list of device files.

Udev

udev provides a dynamic device directory and creates or removes device files automatically when devices are connected or removed.

C) Performance Monitor

Performance monitoring tools help administrators monitor and debug system performance issues.

Some important performance monitoring commands are:

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1. vmstat

vmstat displays information about:

- processes
- memory
- paging
- CPU activity
- I/O activity

Command

vmstat 3

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2. top

top displays real-time information about running processes.

It shows:

- CPU usage
- memory usage
- active processes

Command

top

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3. free

free command displays:

- total memory
- used memory
- free memory
- swap memory

Command

free

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4. iostat

iostat displays CPU statistics and disk I/O statistics.

Command

iostat

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5. netstat

netstat displays:

- network connections
- routing tables
- interface statistics

Command

netstat -tulpn

Q. Differentiate between Multi programmed and Multitasking operating system. [4 m]

Ans.

Multiprogramming OS	Multitasking OS
Multiple programs are kept in memory at the same time.	Multiple tasks run at the same time.
CPU works on one program after another.	CPU switches quickly between tasks using time sharing.
Main aim is to keep CPU busy.	Main aim is to provide smooth user interaction.
User interaction is less.	User interaction is more.
Process changes when current process waits for I/O.	Process changes after fixed time interval.
Execution speed is slower compared to multitasking.	Execution speed and response are faster.
Mainly used in batch systems.	Used in modern operating systems like Windows and Linux.
Ex:- Batch processing systems.	Ex:- Windows, Linux, Android.

Q. Explain Time sharing O.S. [4 m]

Ans.

- In Time Sharing Operating System, the CPU executes multiple jobs by switching among them rapidly.
- The switching occurs so frequently that users can interact with each program while it is running.
- It is an interactive operating system which provides direct communication between the user and the system.

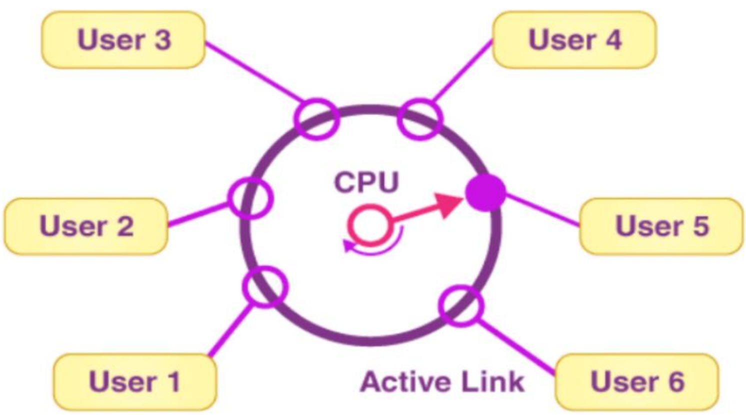
- A time-sharing system allows many users to share computer resources simultaneously.
- In this system, CPU time is divided among users on a scheduled basis.
- The operating system allocates a small amount of CPU time called **time slice** or **time quantum** to each user/process.
- When the time slice of one process expires, the CPU switches to the next process.
- Since switching is very fast, every user feels that they are using the CPU alone.
- The main objective of time-sharing operating system is to:
    - minimize response time
    - provide quick processing
    - improve CPU utilization

Features of Time Sharing OS

- Supports multiple users
- Fast CPU switching
- Better CPU utilization
- Interactive system
- Quick response time

Examples

- UNIX
- Linux



Q. Describe any two components of O.S. [4 m]

Ans.

List of System Components:

1. Process management

2. Main memory management

3. File management

4. I/O system management

5. Secondary storage management

1) Process Management

- A program is a set of instructions.
- When CPU is allocated to a program, it starts execution.
- A program in execution is called a **process**.

Example:

- MS Word running on a computer is a process.
- A process requires various resources such as:
    - CPU time
    - memory
    - files
    - I/O devices
- These resources may be allocated when the process is created or while it is running.

Functions of OS in Process Management

- Creation and deletion of processes
- Suspension and resumption of processes
- Process synchronization

- Process communication
- Deadlock handling

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2) Main Memory Management

- Main memory is a large collection of words or bytes.
- Each location in memory has a unique address.
- It stores data and instructions currently used by CPU.
- CPU directly accesses main memory during program execution.
- Main memory is shared between:
    - CPU
    - processes
    - I/O devices

Functions of OS in Main Memory Management

- Keeping track of used and free memory
- Deciding which process should be loaded into memory
- Allocating memory to processes
- Deallocating memory after process completion

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3) File Management

- A file is a collection of related information stored on secondary storage devices such as:
    - hard disk
    - magnetic tape
    - optical disk
- Files are organized into directories for easy access.

Functions of OS in File Management

- Creation and deletion of files
- Creation and deletion of directories
- Reading and writing files
- Mapping files to secondary storage
- Backup of files

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4) I/O Device Management

- I/O device management handles communication between the system and I/O devices such as:
  - printer
  - scanner
  - keyboard
- Operating system uses **device drivers** to interact with hardware devices.

Functions of OS in I/O Device Management

- Buffering
- Caching
- Spooling
- Device driver management

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5) Secondary Storage Management

- Secondary storage is used to store data permanently.
- It is needed because main memory is temporary and smaller in size.

Examples:

- Hard disk

- SSD
- Tape drive

Functions of OS in Secondary Storage Management

- Free space management
- Storage allocation
- Disk scheduling

Q. Compare between Command Line Interface (CLI) and Graphical User Interface (GUI). [4 m]

Ans.

Parameter	**Command Line Interface (CLI)**	**Graphical User Interface (GUI)**
Definition	Interaction is done by typing commands.	Interaction is done using graphics, icons and visual components.
Understanding	Commands must be memorized.	Icons and visuals are easy to understand.
Memory Requirement	Requires less memory.	Requires more memory because of graphics.
Working Speed	Faster because keyboard commands are used.	Slower because mouse interaction is used.
Resources Used	Mainly keyboard is used.	Both mouse and keyboard are used.
Accuracy	High accuracy.	Comparatively less accuracy.
Flexibility	Interface remains mostly same over time.	Design and structure may change with updates.

Parameter	Command Line Interface (CLI)	Graphical User Interface (GUI)
User Friendly	Less user friendly for beginners.	More user friendly and easy to use.
Examples	DOS, UNIX	Windows, macOS

Q. Write any four system calls related to file management. [4 m]

Ans.

System calls related to file management are:

1. create new file
2. delete existing file
3. open file
4. close file
5. create directories
6. delete directories
7. read, write, reposition in file
8. getfile attributes
9. set file attributes

Q. Write two uses of following O.S. tools

(i) Device Management

(ii) Performance monitor

(iii) Task Scheduler [4 m]

Ans.

i) Device Management

Uses:

- Manages hardware devices like printer, keyboard, scanner, etc.
- Allows communication between operating system and hardware using device drivers.
- Allocates devices to processes when needed.
- Monitors device status and working.

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ii) Performance Monitor

Uses:

- Monitors CPU and memory usage of computer.
- Helps to find system performance problems.
- Checks how programs affect system speed.
- Observes system performance changes.

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iii) Task Scheduler

Uses:

- Assigns CPU to tasks ready for execution.
- Executes tasks automatically at scheduled time.
- Helps in automatic running of background jobs.
- Improves system efficiency.